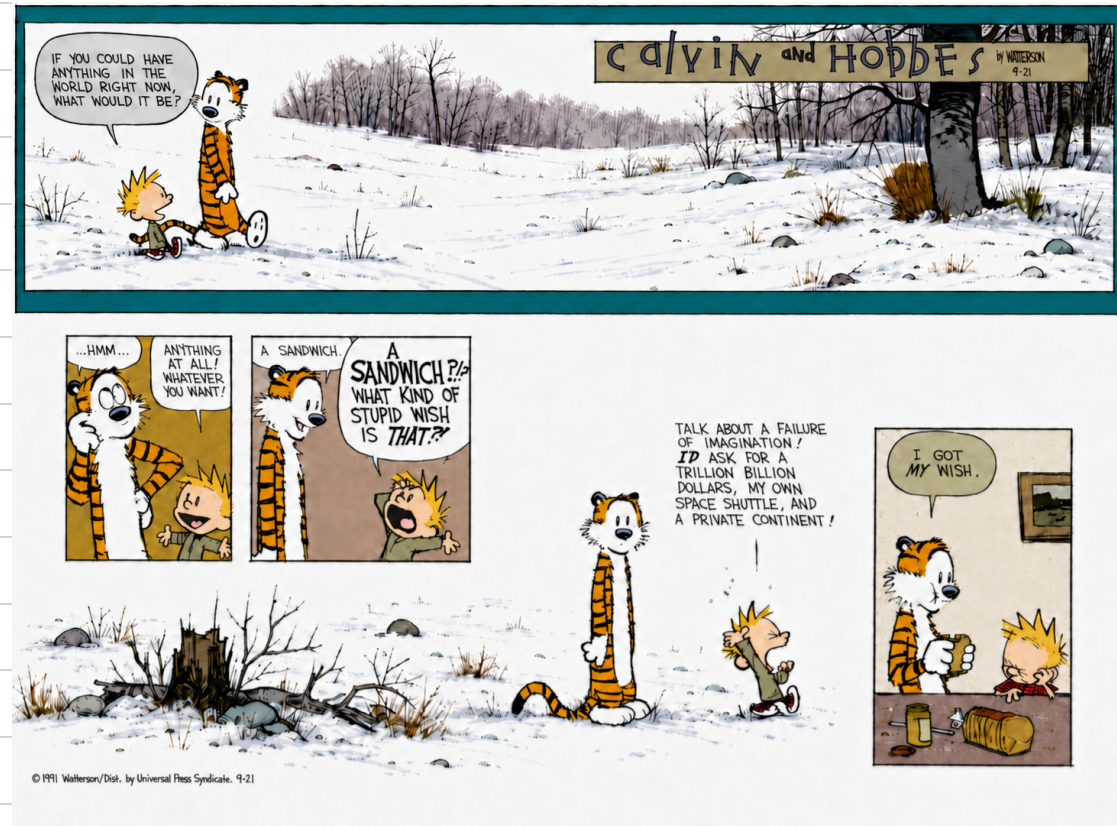




How does reciprocal rank fusion works?

$$RRF(d) = \sum \frac{1}{K + \text{Rank}(d)} \quad \begin{array}{l} K=10 \\ K=11 \\ K=0 \end{array}$$

B1125

D_1

D_2

D_3

D_4

Embedding.

D_2

D_4

D_5

D_1

$$RRF(D_1) = \frac{1}{10+1} + \frac{1}{10+4} = \frac{1}{11} + \frac{1}{14} = 0.1623$$

$$RRF(D_2) = \frac{1}{10+2} + \frac{1}{10+1} \Rightarrow \frac{1}{12} + \frac{1}{11} = 0.1742$$

$$D_3 = 0.0761$$

$$D_4 = 0.1547$$

$$D_5 = 0.0769$$

D_2

D_1

$D_4 \quad D_3 \quad D_5$